

AI Signal — Technical Architecture

AI Orbit / GraphOne unified intelligence ingestion and discovery platform

1. Architecture

Sources → Discovery → Async Extraction → Cleaning/Normalization
→ Deduplication → Entity Resolution → Optional LLM Extraction
→ Relationship Mapping → Validation → SQLite/JSON → Public Explorer

The unified implementation combines the high-volume source collectors, SQLite/WAL storage, RapidFuzz resolution, LLM fallback, and Google Sheets output from the supplied GraphOne project with the canonical entity model, incremental processing, MCP/repository/model coverage, evidence-backed relationships, and public explorer from the AI Orbit project.

2. Incremental ingestion

Source URLs are the primary idempotency key. A normal run discovers current source items but persists only unseen URLs. Existing entities are not reprocessed. Dynamic metrics such as GitHub stars are refreshed separately so normal ingestion remains deterministic and inexpensive.

```
known URL? ■■ yes → skip
            ■
            no
            ↓
extract → normalize → resolve → validate → persist
```

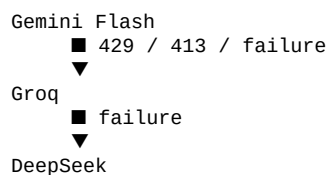
3. Data and graph integrity

Every canonical entity has a stable UUID, entity type, name, URL, categories and source. Specialized metadata is retained for models, repositories, MCP servers, companies, research papers, jobs and news. Relationships contain source/target IDs, relation type, evidence URL, confidence and a deterministic reason.

Supported graph edges include DEVELOPS, SOLVES, INTEGRATES_WITH, MAINTAINS, HAS_REPOSITORY and RUNS.

4. LLM orchestration and resilience

The LLM layer is optional and operates only on source text. Gemini Flash is the primary provider, followed by Groq and DeepSeek. Payloads are chunked before requests. 429 responses use exponential backoff and jitter; 413/context errors cause a smaller chunk or fallback. Authentication failures disable only the affected provider.



The LLM cannot create an entity without a legitimate originating URL. This prevents hallucinated records from entering the canonical dataset.

5. Scale strategy

The trial uses SQLite/WAL for zero-configuration transactional persistence. For 500k+ entities, the same source-URL/stable-ID contract can move to PostgreSQL. Crawling can be partitioned by source/domain across workers, Redis can coordinate distributed rate limits and leases, object storage can retain raw HTML/RSS, and a graph database can support high-volume relationship traversal. A vector store remains an optional retrieval index, never the identity source of truth.

6. Freshness and anti-bot policy

News and jobs are accepted into the fresh dataset only when their normalized publication timestamp is inside the configured 24-hour window. Undated content is not silently considered fresh. API and RSS sources are preferred over scraping. 403/CAPTCHA responses are treated as a stop signal; permitted browser rendering is a fallback for JS-heavy sources.

7. Public explorer

The root static application provides search, category/source/type filters, pagination, entity details, GitHub metrics, connected entities and relationship evidence. It requires no login and contains no API credentials, making it suitable for public Vercel deployment.